

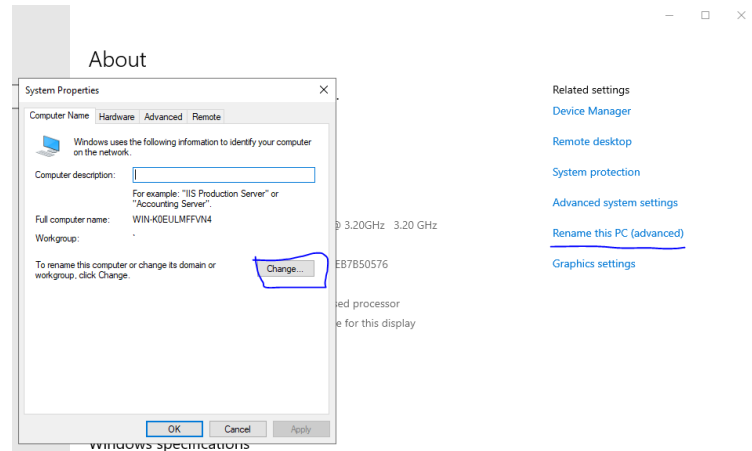
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CNSA-161

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DNS without ADDS

1. Start with removing the server 2022 Dell from the CNSA.LAB domain, go to settings, system, about, and click rename this PC (advanced), click change, as seen in the image on the right, then type anything into the box that says workgroup.



IPv4

☒ On

IP address

10.200.100.3

Subnet mask

255.255.0.0

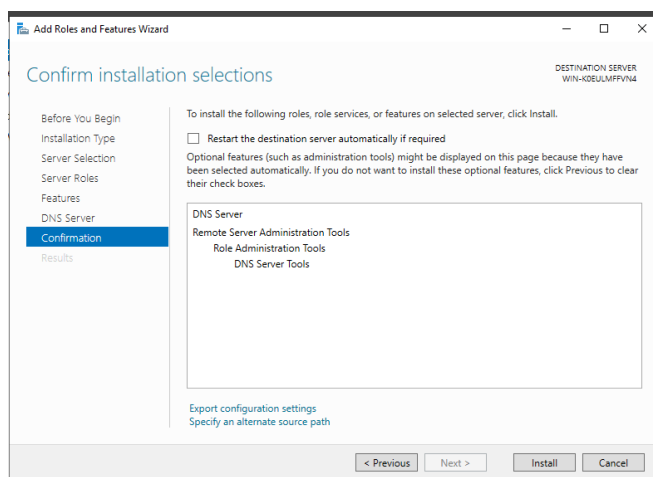
Gateway

10.200.0.1

Preferred DNS

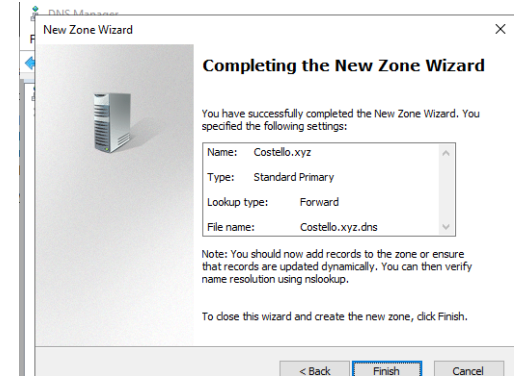
10.200.100.3

2. To statically assign it's IP address, go to settings, ethernet, click on the adapter, and click edit IP settings, set to 10.200.100.3, set subnet mask to 255.0.0.0 set gateway to 10.200.0.1, set preferred DNS server to 10.200.100.3, as seen in the image to the left.



3. Installing DNS just takes going to server manager and clicking add roles and features, in the server roles tab of the wizard that pops up check the box for DNS server and then click install, it should look like the image to the left right before hitting the install button.

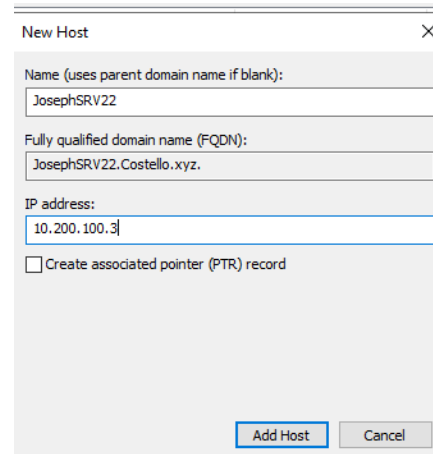
4. Set up the forward lookup zone by selecting the tools button in server manager and clicking DNS, right click the PC name in DNS manager and set it as a primary, forward lookup, see image on right for clarification.



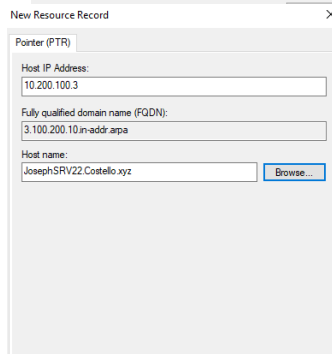
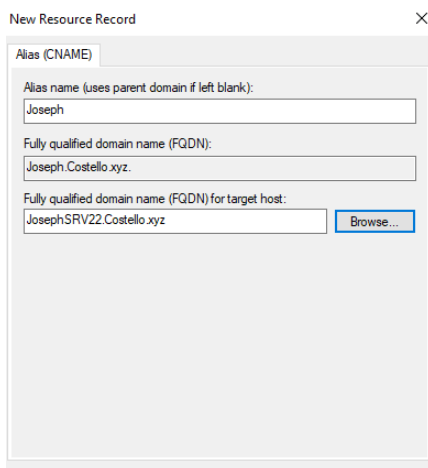
5. The SOA records are to define the hierarchy of the zone administration such as which is the primary and the NS records identify which DNS servers are authoritative for the domain, picture on the left.

Name	Type	Data
(same as parent folder)	Start of Authority (SOA)	[1], win-k0eulmffvn4, hos...
(same as parent folder)	Name Server (NS)	win-k0eulmffvn4.

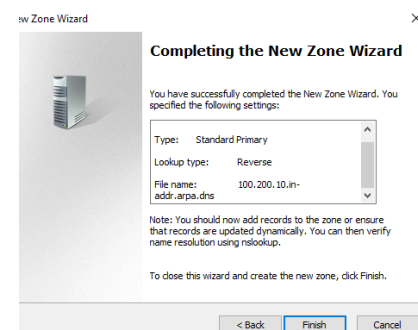
6. Creating an A record is pretty simple, just having to right click on the folder named lastname.xyz, and clicking on 'new host (A or AAAA)' then put these options into the box that comes up. Image on left



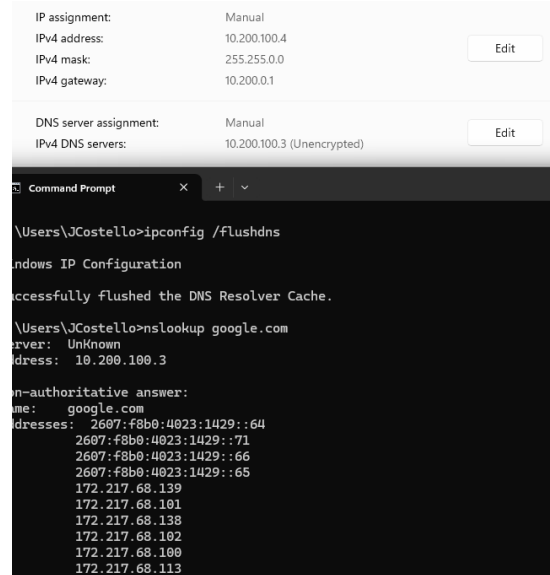
7. Creating a CNAME is the same process just clicking where it says 'new alias (CNAME)' and putting these settings in. Image on right.



8. To create the PTR record, you will right click on reverse lookup zones and click new zone, set it as a primary zone, ipv4, and put the network id as 10.200.100. Then in the reverse lookup zone right click and choose new pointer, setting the host IP address to the windows server 2022 IP address and the host name to the same as the CNAME and A record things. Images on right and left.

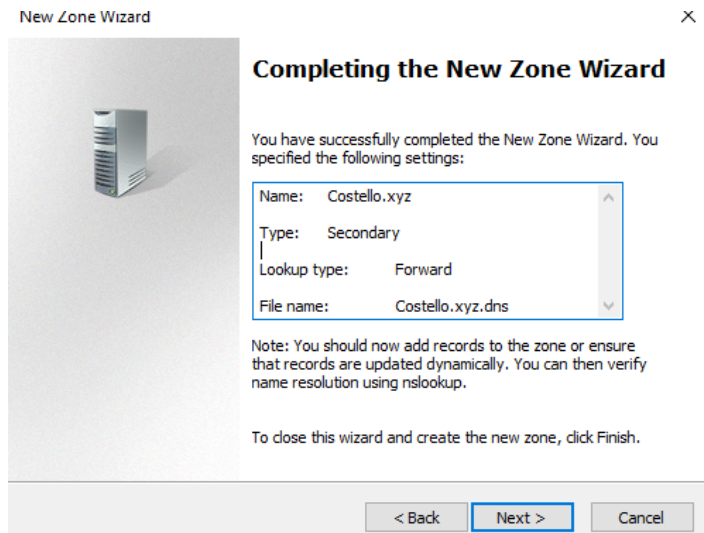
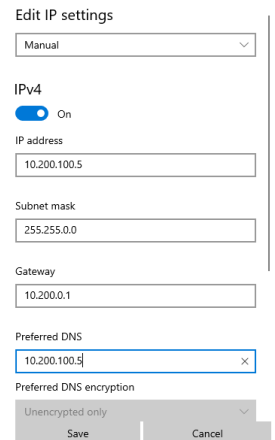


9. To point your windows 11 to use the DNS you just set up on the Server, go into settings, 'network & internet', 'ethernet', and set your IP address to another IP in the same subnet as your server, set your subnet mask and default gateway to the same as your server and set your DNS server to the IP address of your server. Image on the right.



10. Then test that your DNS server works through the server by going into command line and putting in 'ipconfig /flushdns' 'nslookup google.com' This will make sure that your PC doesn't get the connection from it's own cache and you should get a list of address it bounced through to reach google.com. Image on the right

11. To set up the VM as a secondary server start by installing DNS onto it the same way you did on the now VM server. Make sure the VM is set up with the correct subnet mask and default gateway, matching the Server 2022. Images on the right and left



12. Then launch the DNS tool, expand the folder for the PC and right click the forward lookup zone folder, click new zone then in the wizard select secondary zone and name it the same as the zone we set on the windows 22 machine. When it asks for the DNS server you want to copy the zone from put the IP address of your windows 22 machine.

Allow Zone Transfers

DNS zone:
Costello.xyz

IP addresses of the secondary servers:

IP Address	Server FQDN	Validated
10.200.100.3		

13. On both PCs, right click in the forward lookup zone and select properties, go to the zone transfers window and put only to the following servers then click the edit button and put in the other server's IP address as seen in the image on the left.

14. Check that it works by going back to the windows 11 PC and changing it's DNS server to the IP address from the VM, then run the same commands as in step 10 in command line, like in the image on the left.

```
C:\Users\JCostello>nslookup google.com
Server:      UnKnown
Address:     10.200.100.5

Non-authoritative answer:
Name:        google.com
Addresses:   2607:f8b0:4023:1421::65
             2607:f8b0:4023:1421::66
             2607:f8b0:4023:1421::64
             2607:f8b0:4023:1421::8b
             192.178.153.113
             192.178.153.102
             192.178.153.138
             192.178.153.100
             192.178.153.101
             192.178.153.139
```